

NEW T&M DEVICES

Software-defined instrumentation, parallel processing technologies and new methods for wireless and semiconductor test will significantly improve the efficiency of test and measurement systems in 2009



Keithley 4200 semiconductor characterisation system

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The need for test and measurement (T&M) is across industries. Whether in automotive or manufacturing plants, engineers and scientists need flexible, cost-effective solutions for test, control and design to ensure quality, safety and the most advanced products.

T&M devices are required during the lifecycle of any product, beginning from the design phase to the development phase and later in the production phase. For example, while designing a receiver module, a signal generator is used to provide the necessary signals in order to stimulate a response from the receiver. A spectrum analyser is often used in this setup as the refer-

ence receiver, and the results from the test receiver and spectrum analyser are compared. Using these results the design of the test receiver is constantly modified.

One major challenge faced by test engineers is to stay ahead of the technology curve. That is, in order to test the product it is imperative for them to have extensive knowledge of the subject. Given the transient nature of technology, it becomes a real challenge for the test engineers to keep abreast with the new developments and address these applications in an ever-expanding domain.

Major applications in this digital age

Radical advances in digital technology are forging a globally networked society,

bringing huge changes to the electronics industry. Computers, consumer electronics and mobile devices are merging into one always-connected, networked realm, changing the way we communicate, work, play and travel by reshaping our relationship with the world around us.

There are five major application areas in the new digital age for T&M devices: digital RF, high-speed serial data, digital video technologies, embedded system technologies and next-generation network.

Challenges before engineers

The digital-age applications pose many challenges to test engineers. One major challenge is to make the underlying complexity of the next-generation technologies appear simple. This can be achieved through improved functionality, ease of use, greater performance and seamless interoperability.

Digital RF challenge. Digital RF introduces new problems, including runt pulses and broadband transients that cannot be seen with conventional tools. The time spent troubleshooting these problems has a direct impact on the products' time to market. These problems are often DSP-related, and may not be easily repeatable. If the problem only occurs once a day, being able to capture the state of the device-under-test (DUT) the first time it happens can literally take days to find the root cause.

Engineers need to find spurious signals across a broad frequency range, and these signals are getting smaller and moving faster than ever before. Once the signals are found, they need